Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21322

F-422

B.C.A. (Fifth Semester)

Examination, 2025-26

(New Course)

Paper - BCA-502-N

Computer Network

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry **equal marks** i.e. 14 marks symbols used have their usual meanings.

1. (a) How many types of computer networking? Explain each with their architecture.

   (b) Describe the connection oriented and connection less services in detail with an example.

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2. (a) How many layers in OSI reference model? Explain the function of each layer.

   (b) How many types of transmission media? Explain it with an example.

3. (a) Describe the message switching and packet circuit switching with an example.

   (b) How many types of error detection method? Describe any one with an example.

4. (a) Describe the Sliding Window Protocol with an example.

   (b) Differentiate between ALOHA and CSMA multiple access protocol.

5. (a) Explain the Static and Dynamic channel allocation in LANs and MANs with an example.

   (b) Differentiate between Source Routing Bridges, Gateways and Routers.

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6. (a) Describe the congestion control and prevention policies in details.

   (b) What are Network Layer design issue? Explain it in details?

7. (a) How many types of IP address? Explain it with an example.

   (b) What is Domain Name System? Explain the all types of Domain Name System with an example.

8. Write short note on any **four** of the following :

   (a) TCP

   (b) Email: Architecture and Services

   (c) TDM and FDM

   (d) Network Security

   (e) CSMA/CD protocals

   (f) File Transfer Protocal (FTP)

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